

DIGITAL DETECTIVE

WEEK 7 PROJECT REPORT

Name of Experiment

Development of Multiple Case, Hint, Store, Audio and Visual Enhancement System in Digital Detective

Objective

The main objective of Week 7 was to improve the Digital Detective game by adding more interactive and engaging features. The existing investigation system was extended with multiple crime cases, a hint-based assistant, a detective store, scoring and coin systems, sound effects, and an improved user interface. These features make the game more interesting and provide the player with more choices during investigation.

Introduction

Digital Detective is a Java-based investigation game developed using Java Swing and AWT. In the previous weeks, the basic detective character, investigation system, clues, NPC interaction, puzzle, enemy and RPG-related features were developed. In Week 7, the main focus was to make the game feel more like a complete detective game.

The most important improvement was the introduction of multiple cases. Instead of having only one investigation, the player can now select from different crime cases. Each case has its own title and different types of evidence. A hint system was also introduced to help the player find clues when they are having difficulty. A store system was added where the player can use collected coins to purchase hints or increase their score.

Features Implemented

1. Multiple Case System

A multiple case system was added to provide different investigation scenarios. The player can select a case from the main menu by pressing **1, 2, or 3**.

The available cases are:

- **The Missing Diamond**
- **The Stolen Laptop**
- **The Secret Document**

Each case contains five different clues. When the player finds a clue, it is added to the evidence list and the corresponding score is added. This makes the investigation system more flexible and gives the game a basic storyline-based structure.

2. Hint-Based Assistant

A hint system was developed to assist the detective during investigation. The player can press **H** to request a hint. The detective assistant checks the current investigation progress and provides useful information, such as suggesting an area where an unfound clue may be located.

Hints require coins, which adds an additional challenge to the game. The player must decide when a hint is necessary instead of using it continuously.

3. Store System

A simple detective store was introduced using the **B** key. The store uses coins collected during the investigation.

The player can purchase:

- **Hint – 20 Coins**
- **Score Boost – 30 Coins**

When a clue is collected, the player receives additional coins. Coins can therefore be saved and used later when the player needs assistance or wants to increase the score.

4. Scoring System

The scoring system was improved to make the investigation more rewarding. Different clues provide different amounts of score depending on their importance. Talking to the NPC and solving the investigation puzzle also provides additional points.

After all required investigation tasks are completed, a bonus score is awarded. The final score is displayed on the case completion screen.

5. Audio Enhancement

Basic sound feedback was added to the game. A Java system beep is used when important actions occur, such as starting a case, collecting evidence, purchasing a score boost, or completing the investigation.

This provides immediate audio feedback without requiring additional external sound files.

6. Visual and UI Enhancement

The user interface was improved to make important game information easier to understand. The game screen now displays the current case name, score, coins and collected evidence count.

Different colours and shapes are used to identify important objects. For example, the detective is represented by a blue character, clues are shown as yellow objects, the NPC is represented by a red object, and the puzzle area is represented by an orange object.

The main menu was also improved so that the player can select a case before starting the investigation.

Working Procedure

When the game starts, the main menu is displayed. The player selects one of the available cases and presses **ENTER** to begin.

After starting the case, the detective can move around the investigation area using the **WASD keys or arrow keys**. When the player reaches a clue, pressing **E** collects the evidence and increases the score and coins.

The player can interact with the NPC by approaching the red NPC area and pressing **E**. The NPC provides information related to the investigation.

An investigation puzzle is also included. The player must investigate the puzzle area and enter the correct answer. The current puzzle accepts **SUSPECT** as the correct answer.

During the investigation, the player can press **H** to receive a hint or **B** to open the store. After collecting all evidence, talking to the NPC and successfully completing the puzzle, the case is marked as solved.

Result

The Week 7 development successfully extended the Digital Detective game with multiple cases, a hint assistant, a store and coin system, improved scoring, audio feedback and a better user interface. The game now provides more interaction and gives the player additional objectives during investigation.

The completed system makes the project more organized and closer to a complete detective-style game. Players can select different cases, collect evidence, use resources, receive assistance and finally solve the investigation.

Discussion

The implementation of Week 7 improved both the gameplay and presentation of Digital Detective. The multiple case feature makes the game more expandable because additional cases can be added in the future. The hint and store systems introduce a basic resource-management element, while the scoring system gives the player a reason to explore the investigation area carefully.

The improved interface also makes important information such as score, coins and evidence progress visible during gameplay. Audio feedback provides additional interaction when important events occur.

Overall, Week 7 completed an important stage of the Digital Detective project by combining the investigation mechanics developed in previous weeks with additional gameplay and presentation features.

